Regulatory Toxicology and Pharmacology 156 (2025) 105766

Contents lists available at ScienceDirect

Regulatory Toxicology and Pharmacology

journal homepage: www.elsevier.com/locate/yrtph

Evaluating dermal absorption of perfluorooctanoic acid (PFOA) and implications for other per- and polyfluoroalkyl substances (PFAS)

Andrew Yeh [a,*], Robyn L. Prueitt [a], Laura E. Kerper [b], Barbara D. Beck [b]

[a] Gradient, 600 Stewart Street, Suite 1900, Seattle, WA, 98101, USA
[b] Gradient, One Beacon St., 17th Floor, Boston, MA, 02108, USA

ARTICLE INFO

Handling Editor: Dr. Martin Van den berg

Keywords:
Perfluorooctanoic acid (PFOA)
Per- and polyfluoroalkyl substances (PFAS)
Fractional absorption
Permeability coefficient
Steady-state flux
Dimensionless dermal number (N_{derm})
Dermal absorption factor
Skin
Soil
Water

ABSTRACT

To date, only four studies directly measured dermal absorption kinetics of perfluorooctanoic acid (PFOA) in human skin. Reported kinetic parameters spanned two to five orders of magnitude, demonstrating the need to determine the causes of variability and identify the most appropriate dermal absorption factors for use in exposure assessments. We evaluated the reliability and physiological relevance of studies that measured PFOA fractional absorption, steady-state flux (J_{ss}), and dermal permeability coefficient (K_p). We verified whether the reported kinetic parameters were measured under appropriate conditions (*i.e.*, fractional absorption under finite dose conditions, and J_{ss} and K_p under infinite dose conditions). We recommend the following values measured at the approximate pH of the skin, and in aqueous solvents or relevant consumer product matrices, for use as provisional values in PFOA exposure assessments: 1.6% fractional absorption under finite dose conditions, and 0.132 μg/cm^2-hr and 0.000044 cm/h for J_{ss} and K_p, respectively, under infinite dose conditions. Using the recommended absorption factors, we estimated PFOA exposures in children from soil and water *via* dermal and ingestion routes. Our results indicate low dermal absorption of PFOA relative to ingestion, and low dermal absorption is expected for per- and polyfluoroalkyl substances (PFAS) with physicochemical properties similar to PFOA.

1. Introduction

Dermal absorption of per- and polyfluoroalkyl substances (PFAS) is considered a minor exposure pathway for humans relative to ingestion and inhalation (Haug et al., 2011; Poothong et al., 2020). However, there is concern regarding potential dermal exposure to PFAS due to their presence in consumer products (*e.g.*, textiles, cosmetics, personal care products). Few studies have evaluated dermal exposure to PFAS, and the majority focused on perfluorooctanoic acid (PFOA) and perfluorooctane sulfonate (PFOS). In addition, most of these studies did not measure the rate and extent of PFAS dermal uptake that occurred under the exposure conditions. Accordingly, little is known regarding the dermal absorption kinetics of PFAS (*e.g.*, Ragnarsdóttir et al., 2022, 2023, 2024; Kissel et al., 2023).

Scientific and regulatory agencies often estimate dermally absorbed doses of chemicals using a fixed fractional absorption approach. An experimental measurement of fractional absorption (*i.e.*, the fraction of the applied chemical absorbed through the skin), when available, is applied as a dermal absorption factor. In such assessments, the dermally absorbed dose of a chemical is related to its fractional absorption and

Abbreviations: 3D-HSE, 3D-Human Skin Equivalent; ABS_d, Dermal Absorption Factor; ADD_{Soil}, Dose of PFOA from Ingestion of Soil; ADD_{Water}, Dose of PFOA from Ingestion of Water; AF, Soil-to-Skin Adherence Factor; APFO, Ammonium Salt of Perfluorooctanoic Acid; ATSDR, Agency for Toxic Substances and Disease Registry; CF, Conversion Factor; C_{Soil}, Concentration of PFOA in Soil; C_v, Concentration of PFOA in the Vehicle; C_{Water}, Concentration of PFOA in Water; DAD_{Soil}, Dermally Absorbed Dose of PFOA from Soil; DAD_{Water}, Dermally Absorbed Dose of PFOA from Water; DA_{Soil}, Dose of PFOA Absorbed in Skin at the End of Soil Exposure; DA_{Water}, Dose of PFOA Absorbed in Skin at the End of Bathing; DDT, Dichloro-diphenyl-trichloroethane; ED, Experimental Duration; EL, Experimental Load; HEM, Human Epidermal Membrane; HFT, Full-Thickness Human Skin; HSE, Human Skin Equivalent; J_{ss}, Steady-State Flux; K_{ow}, Octanol-Water Partition Coefficient; K_p, Dermal Permeability Coefficient; MRL, Minimum Reporting Limit; N_{derm}, Dermal Number; PFAS, Per- and Polyfluoroalkyl Substances; PFCA, Perfluoroalkyl Carboxylic Acid; PFOA, Perfluorooctanoic Acid; PFOS, Perfluorooctane Sulfonate; PFSA, Perfluorosulfonic Acid; pKa, Acid Dissociation Constant; ppt, Parts per Trillion; SCCS, Scientific Committee of Consumer Safety; TCDD, 2,3,7,8-Tetrachlorodibenzo-p-dioxin; UCMR 3, Third Unregulated Contaminant Monitoring Rule; UCMR 5, Fifth Unregulated Contaminant Monitoring Rule; US, United States; US EPA, United States Environmental Protection Agency.

* Corresponding author.
E-mail addresses: andrew.yeh@gradientcorp.com (A. Yeh), robyn.prueitt@gradientcorp.com (R.L. Prueitt), barbara.beck@gradientcorp.com (B.D. Beck).

https://doi.org/10.1016/j.yrtph.2024.105766
Received 21 July 2024; Received in revised form 8 December 2024; Accepted 16 December 2024
Available online 17 December 2024
0273-2300/© 2024 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

other factors such as exposure media (*e.g.*, soil, water), duration and frequency of exposure, and loading concentration (US EPA, 2004). For PFAS, dermally absorbed doses have been estimated using a fractional absorption approach, but only four studies have directly measured PFAS fractional absorption. Three of these studies focused on PFOA (Franko et al., 2012; Abraham and Monien, 2022) or the ammonium salt of perfluorooctanoic acid (APFO) (Fasano et al., 2005), and the remaining study evaluated PFOA and 16 other PFAS (10 perfluoroalkyl carboxylic acids [PFCAs] and 7 perfluorosulfonic acids [PFSAs] total) (Ragnarsdóttir et al., 2024). Measurements of PFOA fractional absorption in these studies ranged by three orders of magnitude, 0.048% (Fasano et al., 2005), 1.6% (Abraham and Monien, 2022), 13.5% (Ragnarsdóttir et al., 2024), and 48–69% (Franko et al., 2012). The wide variability of the reported measurements demonstrates the need to identify the most appropriate absorption factors for use in PFOA dermal exposure assessments (and for other PFAS with similar chemistry). For example, Poothong et al. (2020) estimated dermal absorption of PFAS using a dermal absorption factor of 0.48, based on the results of Franko et al. (2012), but the authors stated this value was likely an overestimate.

Here, we conducted a critical evaluation of studies that performed direct experimental measurements of PFOA dermal absorption kinetics in human skin. The first objective of our evaluation was to identify the most reliable and physiologically relevant measurements of PFOA dermal absorption kinetic parameters for use in human exposure assessments: fractional absorption, steady-state flux (J_{ss}), and dermal permeability coefficient (K_p). The reliability of the kinetic measurements was determined primarily based on whether the experiments were conducted under the appropriate dose conditions (*i.e.*, fractional absorption under finite dose conditions, and J_{ss} and K_p under infinite dose conditions). The physiological relevance of the kinetic measurements (*i.e.*, their relevance to human exposures) was based on evaluation of other experimental conditions (*i.e.*, vehicle solvent, pH). The second objective of our evaluation was to apply the identified absorption factors to estimate dermally absorbed doses of PFOA in two hypothetical exposure scenarios in children exposed *via* soil or bathing, and compare these estimates to respective exposures *via* ingestion.

2. Methods

2.1. Identifying PFOA dermal absorption kinetic studies in human skin

We searched the PubMed database using the terms "dermal," "perfluorooctanoic acid," and "PFOA." The search encompassed literature published through July 2024. We also identified PFOA dermal exposure studies reviewed in toxicological reviews of PFAS published by scientific and regulatory agencies (US EPA, 2016; ATSDR, 2021). We focused only on studies conducted on human skin, either *in vitro* (*e.g.*, human epidermal membranes [HEMs]) or *in vivo*. We identified four studies that measured fractional absorption, J_{ss}, and/or K_p (Fasano et al., 2005; Franko et al., 2012; Abraham and Monien, 2022; Ragnarsdóttir et al., 2024). Fasano et al. (2005) and Franko et al. (2012) conducted experiments in human and rat skin *in vitro* models, but only the human *in vitro* data were evaluated here (full-thickness or heat-separated HEMs from donor subjects). Ragnarsdóttir et al. (2024) conducted experiments in an *in vitro* model (3D-human skin equivalent [3D-HSE]). Abraham and Monien (2022) conducted an *in vivo* exposure experiment in one human male subject.

Due to the availability of experimental measurements of dermal absorption kinetics in human skin, we did not evaluate analogous data from rodent skin studies. Briefly, we identified three studies that directly measured PFOA dermal absorption kinetics in rodent skin, two in rats (Fasano et al., 2005; Chen et al., 2022) and one in mice (Franko et al., 2012). Similar to results in humans, the available measurements of PFOA dermal absorption kinetics in rodents are variable. For example, measurements of fractional absorption ranged from 1.44% (Fasano et al., 2005) to 6.9–10.1% in rat skin (Chen et al., 2022), and 50% in mouse skin (Franko et al., 2012). Notably, in the *in vitro* experiments conducted by Fasano et al. (2005), which reported the lowest estimates of PFOA dermal fractional absorption for both rodents and humans, fractional absorption was 30-fold greater in rat skin compared to human skin (*i.e.*, 1.44 and 0.048%, respectively). These results are consistent with the understanding that dermal absorption of chemicals is greater in rodents relative to humans and that extrapolation of dermal exposure from rodent models can lead to overestimates of dermal exposure in human health risk assessment (OECD, 2022).

2.2. Assessing the reliability and physiological relevance of experimental measurements of PFOA dermal absorption kinetics

Dermal absorption experiments can be conducted under so-called finite or infinite dose conditions, and certain kinetic parameters are more reliably measured under finite *versus* infinite conditions (Table 1). Finite dose experiments involve low doses and substantial depletion of the applied load over the exposure period, allowing for reliable measurement of fractional absorption relevant to realistic environmental or occupational exposures (Kissel, 2011; Frasch et al., 2014; OECD, 2022), and are relevant to dermal absorption from solid matrices, such as soil, on skin. However, finite dose conditions do not reliably measure other dermal absorption kinetic parameters, such as J_{ss} (*i.e.*, the steady-state absorption rate). For most chemicals, flux varies inversely with dose, and therefore, due to dose depletion over the exposure period, experimental flux will be less than the maximum J_{ss} of the chemical (Kissel, 2011; Frasch et al., 2014).

Infinite dose experiments involve sufficiently high doses that ensure minimal load depletion over the exposure period, allowing for reliable measurement of J_{ss} because the lack of substantial modulation of the applied load allows experimental flux to approach J_{ss} (Kissel, 2011; Frasch et al., 2014). Infinite dose conditions are also best suited for measurement of K_p (the rate of penetration of a chemical across skin) and are relevant to dermal absorption from liquid matrices, such as water. If J_{ss} is not experimentally measured, it may be estimated based on K_p and the chemical's concentration (Frasch et al., 2014). In contrast, infinite dose conditions do not reliably measure fractional absorption because, for many compounds, fractional absorption varies inversely with the applied dose. Thus, the excessive dose results in a low percentage of absorption that does not represent the true absorption potential of the chemical (Kissel, 2011; Frasch et al., 2014; OECD, 2022). In addition, the high doses are not representative of lower, more realistic doses relevant to environmental or occupational exposures.

The dermal number (N_{derm}, *i.e.*, the dimensionless ratio of mass delivery to skin to potential absorptive flux) can be calculated to assess the reliability of measurements of dermal absorption kinetics (Kissel, 2011). Low values of N_{derm} indicate that the study was conducted under finite dose conditions that are appropriate for reliable measurement of fractional absorption and are likely representative of dermal exposures expected in environmental or occupational settings (Frasch et al., 2014). High values of N_{derm} indicate that the study was conducted under infinite dose conditions that are appropriate for reliable measurement of

Table 1
Characteristics of finite and infinite dose experimental conditions.

Finite Dose Conditions	Infinite Dose Conditions
Load depletion expected to occur	Load is not likely to be depleted over time
Peak flux is less than maximum	Reliable measurement of J_{ss} and K_p
Reliable measurement of fractional absorption relevant to realistic environmental or occupational exposures	Low fractional absorption relative to applied load, not representative of true absorption potential
Low values of N_{derm}	High values of N_{derm}

Notes.
J_{ss} = Steady-State Flux; N_{derm} = Dermal Number.

Jss and K_p, but are not representative of typical environmental or occupational exposures. As shown in Equation (1), N_{derm} was calculated for each kinetic study when the values required for calculation were reported by the study authors. If J_{ss} was not reported, it was estimated as shown in Equation (2) based on the experimentally determined K_p and the concentration of PFOA in the vehicle (C_v).

Equation 1		$N_{derm} = \frac{EL}{J_{ss} \times ED}$	
Parameter		Definition (Unit)	Value
N_{derm}	=	Dermal number (unitless)	–
EL	=	Experimental load (μg/cm^2)	Experimental value
J_{ss}	=	Steady-state flux (μg/cm^2 × hrs)	Experimental or estimated value
ED	=	Experimental duration (hrs) Experimental value	
Equation 2		$J_{ss} = K_p \times C_v$	
Parameter		Definition (Unit)	Value
J_{ss}	=	Steady-state flux (μg/cm^2 × hrs)	–
K_p	=	Dermal permeability coefficient (cm/hrs)	Experimental value
C_v	=	Concentration in vehicle (μg/cm^3)	Experimental value

Here, we considered kinetic measurements to be reliable based on whether the experiments were conducted under the appropriate experimental conditions, specifically fractional absorption under finite dose conditions, and J_{ss} and K_p under infinite dose conditions. Our evaluation of reliability involved the calculation of N_{derm}, which is not a standardized method to assess data reliability. Thus, our definition of reliability differs from those employed in standardized guidelines (*e.g.*, European Commission Health & Consumer Protection Directorate-General Scientific Committee on Consumer Safety [EC SCCS], 2021; EFSA, 2017; OECD, 2022). Consistent with the European Union's Scientific Committee on Consumer Safety (EC SCCS) guidelines, however, we also considered that reliable dermal absorption studies should evaluate at least eight skin samples from at least four subjects (EC SCCS, 2021). Finally, our evaluation of physiological relevance (*i.e.*, relevance to human exposures) was based on the vehicle solvent and pH employed in the experiments.

2.3. Estimating PFOA dermal absorption in hypothetical exposure scenarios

Experimentally derived fractional absorption and K_p values can be applied as parameters in quantitative exposure assessments for soil and water, respectively. We relied on experimentally derived values or values recommended by the United States Environmental Protection Agency (US EPA) for general exposure parameters to evaluate potential dermal exposure risks from PFOA in hypothetical exposure scenarios in children exposed *via* (1) soil, and (2) bathing. Dermal and oral absorption of PFOA was estimated using standard equations and exposure factors for dermal risk assessment of soil and water exposures (US EPA, 2004, 2017a, 2019; Stalcup, 2014).

2.3.1. Scenario 1: dermally absorbed dose of PFOA from soil

The dose of PFOA absorbed in skin at the end of soil exposure (DA_{Soil}) was estimated as shown in Equation (3). The global median soil concentration of PFOA of 0.124 ng/g reported by Strynar et al. (2012), converted to mg/kg (*i.e.*, 0.000124 mg/kg), was used to estimate a representative background concentration of PFOA in soil (C_{Soil}). Strynar et al. (2012) evaluated 60 soil samples from six countries (United States [US], China, Japan, Norway, Greece, and Mexico; 10 samples per country) from areas with no known PFAS contamination and/or that were not in the vicinity of industries known to use PFAS, and noted that the studied soils were intended to be useful indicators of PFOA background concentrations in different soil types and parts of the world. A value of 10^{-6} kg/mg was applied as the conversion factor (CF) (US EPA, 2004). For the weighted soil-to-skin adherence factor (AF), we applied a value of 0.2 mg/cm^2, as recommended by US EPA for child residents (<1 to <6 years old), representing the 95th percentile for children playing at a day care center (central-tendency soil contact activity) as well as the 50th percentile for children playing in wet soil (high-end soil contact activity) (US EPA, 2004; Stalcup, 2014). For the dermal absorption factor (ABS_d), we applied the recommended fractional absorption of PFOA identified as part of our critical evaluation of experimental measurements of PFOA dermal absorption kinetics. The dermally absorbed dose of PFOA from soil (DAD_{Soil}) was calculated as shown in Equation (4). We applied general exposure parameters for children in residential exposure scenarios involving soil contact as recommended by US EPA (Stalcup, 2014).

Equation 3		$DA_{Soil} = C_{Soil} \times CF \times AF \times ABS_d$	
Parameter		Definition (Unit)	Value (Source, if Applicable)
DA_{Soil}	=	Absorbed dose per soil event (mg/cm^2-event)	–
C_{Soil}	=	PFOA concentration in soil (mg/kg) 0.000124	Strynar et al. (2012)
CF	=	Conversion factor (kg/mg)	1^{-6}
AF	=	Soil-to-skin adherence factor (mg/cm^2)	0.2 (Stalcup, 2014)
ABS_d	=	Dermal absorption fraction	Experimental value
Equation 4		$DAD_{Soil} = \frac{DA_{Soil} \times EV \times ED \times EF \times SA}{BW \times AT}$	
Parameter		Definition (Unit)	Value
DAD_{Soil}	=	Dermally absorbed dose from soil (mg/kg-day)	–
DA_{Soil}	=	Absorbed dose per soil event (mg/cm^2-event)	–
EV	=	Event frequency (events/day)	1 (US EPA, 2004)
ED	=	Event duration (yrs)	6 (Stalcup, 2014)
EF	=	Exposure frequency (days/yr)	350 (Stalcup, 2014)
SA	=	Skin surface area (cm^2)	2,373 (Stalcup, 2014)
BW	=	Body weight (kg)	15 (Stalcup, 2014)
AT	=	Averaging time (days)	2,190 (Stalcup, 2014)

2.3.2. Scenario 2: dermally absorbed dose of PFOA from bathing

The dose of PFOA absorbed in skin at the end of bathing (DA_{Water}) was estimated as shown in Equation (5). We applied the recommended K_p identified as part of our critical evaluation of the experimental measurements of PFOA dermal absorption kinetics. To estimate representative PFOA concentrations in water (C_{Water}), we used the median and 90th percentile concentrations of 0.03 and 0.07 μg/L, respectively, reported by the US EPA for PFOA in US drinking water samples as part of monitoring under the third Unregulated Contaminant Monitoring Rule (UCMR 3) from 2013 to 2015 (US EPA, 2017b, 2022). Currently, UCMR 3 is the best available source of national occurrence data for PFOA in drinking water (US EPA, 2022). US EPA reported that detected concentrations of PFOA were above the minimum reporting level (MRL) of 0.02 μg/L, or 20 parts per trillion (ppt), in 379 samples from 117 public water systems (out of 36,972 total samples from 4,920 public water systems), and ranged from 0.02 to 0.349 μg/L; the median and 90th percentile concentrations were 0.03 and 0.07 μg/L, respectively (US EPA, 2017b, 2022). Monitoring for PFOA under the fifth Unregulated Contaminant Monitoring Rule (UCMR 5) is currently underway, but the dataset is incomplete as of the date of publication of the current study. Nevertheless, we expect that our use of UCMR 3 data to estimate PFOA drinking water concentrations will be more conservative than if we used UCMR 5 data. This is because the MRL for PFOA was 20 ppt in UCMR 3, whereas the MRL is 4 ppt in UCMR 5. Accordingly, the median and 90th percentile PFOA concentrations in UCMR 3 are likely to be higher than the respective values to be determined in UCMR 5. The dermally absorbed dose of PFOA from water (DAD_{Water}, reported in mg/kg-day) was calculated as shown in Equation (6). We applied general exposure parameters for children in residential exposure scenarios involving

water contact as recommended by US EPA (Stalcup, 2014).

Equation 5		$DA_{Water} = K_p \times C_{Water} \times T_{Water}$	
Parameter		Definition (Unit)	Value (Source, if Applicable)
DA_{Water}	=	Absorbed dose per water event (mg/cm^2-event)	–
K_p	=	Dermal permeability coefficient (cm/hr)	Experimental value
C_{Water}	=	PFOA concentration in water (mg/cm^3)	3E-8 or 7E-8 (US EPA, 2017b)
T_{Water}	=	Event duration (hrs/event)	0.33 (US EPA, 2004)
Equation 6		$DAD_{Water} = \frac{DA_{Water} \times EV \times ED \times EF \times SA}{BW \times AT}$	
Parameter		Definition (Unit)	Value
DAD_{Water}	=	Dermally absorbed dose from water (mg/kg-day)	–
DA_{Water}	=	Absorbed dose per water event (mg/cm^2-event)	–
EV	=	Event frequency (events/day)	1 (US EPA, 2004)
ED	=	Event duration (yrs)	6 (Stalcup, 2014)
EF	=	Exposure frequency (days/yr)	350 (Stalcup, 2014)
SA	=	Skin surface area (cm^2)	6,365 (Stalcup, 2014)
BW	=	Body weight (kg)	15 (Stalcup, 2014)
AT	=	Averaging time (days)	2,190 (Stalcup, 2014)

2.3.3. Estimating average daily potential dose of PFOA from ingestion of soil and water

The average daily potential doses of PFOA from ingestion of soil (ADD_{Soil}) and water (ADD_{Water}) were estimated as shown in Equations 7 and 8, respectively (US EPA, 2024a). We applied general exposure parameters for children recommended by US EPA (US EPA, 2004, 2017a, 2019; Stalcup, 2014). Accordingly, the ADD_{Soil} was estimated as 1.6×10^{-9} mg/kg-day, and the ADD_{Water} was estimated as 1.7×10^{-6} or 3.9×10^{-6} mg/kg-day based on the median and 95th percentile PFOA water concentrations reported in UCMR 3 monitoring data (US EPA, 2017b, 2022), respectively.

Equation 7		$ADD_{Soil} = \frac{C_{Soil} \times CF \times IRS_C \times EF \times ED}{BW \times AT}$	
Parameter		Definition (Unit)	Value (Source, if Applicable)
ADD_{Soil}	=	Average daily potential dose from soil (mg/kg-day)	–
C_{Soil}	=	PFOA concentration in soil (mg/kg)	0.000124 (Strynar et al., 2012)
CF	=	Conversion factor (kg/mg)	1^{-6}
IRS_c	=	Ingestion rate (mg/day)	200 (Stalcup, 2014)
EF	=	Exposure frequency (days/yr)	350 (Stalcup, 2014)
ED	=	Exposure duration (yrs)	6 (Stalcup, 2014)
BW	=	Body weight (kg)	15 (Stalcup, 2014)
AT	=	Averaging time (days)	2,190 (Stalcup, 2014)
Equation 8		$ADD_{Water} = \frac{C_{Water} \times IRW_C \times EF \times ED}{BW \times AT}$	
Parameter		Definition (Unit)	Value
ADD_{Water}	=	Average daily potential dose from water (mg/kg-day)	–
C_{Water}	=	PFOA concentration in water (mg/mL)	3E-8 or 7E-8 (US EPA, 2017b)
IRW_c	=	Ingestion rate (mL/day)	780 (Stalcup, 2014)
EF	=	Exposure frequency (days/yr)	350 (Stalcup, 2014)
ED	=	Event duration (yr)	6 (Stalcup, 2014)
BW	=	Body weight (kg)	15 (Stalcup, 2014)
AT	=	Averaging time (days)	2,190 (Stalcup, 2014)

3. Results

3.1. Identification of experimental measurements of PFOA dermal absorption kinetics

Table 2 summarizes the experimental conditions and results of the four studies that directly measured the dermal absorption kinetics of PFOA in human skin, *i.e.*, three *in vitro* studies (Fasano et al., 2005; Franko et al., 2012; Ragnarsdóttir et al., 2024) and one *in vivo* study in a human male subject (Abraham and Monien, 2022). We calculated experimental loads (ELs) by dividing the mass of PFOA in the applied load (*i.e.*, μg of PFOA in the volume of solution applied) by the surface area of exposed skin (in cm^2). Calculations of ELs are described in the **Supplemental Information**.

The *in vitro* study by Fasano et al. (2005) employed one experimental condition to measure the parameters of fractional absorption, J_{ss}, and K_p (*i.e.*, APFO dissolved in water at pH 5.5 in HEM). The *in vitro* study by Franko et al. (2012) evaluated five experimental conditions involving PFOA: (1) acetone solvent at unreported pH in full-thickness human skin (HFT), (2) acetone solvent at unreported pH in HEM, (3) water at pH 2.25 in HEM, (4) water at pH 5.01 in HEM, and (5) citrate buffer at pH 5.5 in HEM. Under experimental conditions 1 and 2, the only kinetic parameter measured was fractional absorption. Under experimental conditions 3, 4, and 5, the kinetic parameter measured was K_p. Franko et al. (2012) did not measure J_{ss} under any experimental condition. The *in vivo* study by Abraham and Monien (2022) measured fractional absorption under one experimental condition (*i.e.*, PFOA dissolved in methanol, dried, redissolved in ethanol, and then mixed into sunscreen at unreported pH) and did not measure J_{ss} or K_p. The authors also did not report the pH of the ultimate sunscreen matrix containing PFOA that was applied to the skin. Finally, the *in vitro* study by Ragnarsdóttir et al. (2024) evaluated one experimental condition to measure the parameters of fractional absorption, J_{ss}, and K_p (*i.e.*, PFOA dissolved in methanol at unreported pH in human skin equivalents [HSEs]).

3.2. Calculation of N_{derm}

Table 3 summarizes the approach to calculate N_{derm} values. As discussed, N_{derm} was calculated by applying values for EL, experimental duration (ED), and J_{ss} to Equation (1). The values of EL and ED were available for each study, but J_{ss} values were only available for Fasano et al. (2005) (directly measured); experimental conditions 3, 4, and 5 of Franko et al. (2012) (estimated by applying their measurements of K_p and C_v to Equation (2)); and Ragnarsdóttir et al. (2024) (directly measured). Thus, calculation of N_{derm} values was possible for these five experimental conditions only.

3.3. Selection of the most appropriate experimental measurement of PFOA fractional absorption

Strengths and limitations associated with fractional absorption experimental measurements are summarized in Table 4. We conclude that 1.6% fractional absorption of PFOA as reported by Abraham and Monien (2022) is the most appropriate value for use as a dermal absorption factor in PFOA exposure assessments, given the measurement of absorption under finite dose conditions and from a relevant consumer product matrix (*i.e.*, sunscreen) in a human subject. This value should be considered provisional due to uncertainties associated with the measurement but is the most reliable measurement among the available experimental data.

3.4. Selection of the most appropriate experimental measurements of PFOA J_{ss} and K_p

Strengths and limitations associated with J_{ss} and K_p experimental measurements are summarized in Table 4. We conclude that Fasano et al. (2005) and Franko et al. (2012) (experimental conditions 4 and 5 only) reported equally appropriate measurements of J_{ss} and K_p, given their measurements under infinite dose conditions and dissolved in appropriate vehicle solvents (water or citrate buffer) at physiologically relevant pH (5.01–5.5). Thus, 0.132 μg/cm^2-hr and 0.000044 cm/h for J_{ss} and K_p, respectively, as measured by Franko et al. (2012) provide the most conservative (*i.e.*, protective) estimates of PFOA dermal permeability at pH 5.5.

Table 2
Experimental conditions and results of PFOA dermal absorption studies in human skin.

Study	Test Material	Model	Subjects	Sample Size	Solvent	pH	EL (μg/cm^2)	SA (cm^2)	ED (hrs)	FA (%)	J_{ss} (μg/cm^2-hr)	K_p (cm/hr)
Fasano et al. (2005)	APFO	HEM	≥3	9	Water	5.5	46,900	0.64	48	0.048	0.190	9.5E-7
Franko et al. (2012)	PFOA	Study 1: HFT	1	7	Acetone	NR	781	0.64	24	69.0	–	–
		Study 2: HEM	1	6						47.9	–	–
		Study 3: HEM	3	8	Water	2.25	2,340			–	–	5.5E-2
		Study 4: HEM	3	8		5.01	11			–	–	5.8E-5
		Study 5: HEM	3	10	Buffer	5.5	2,340			–	–	4.4E-5
Abraham and Monien (2022)	PFOA	Human subject	1	1	Sunscreen	NR	0.006	19,652	48	1.6	–	–
Ragnarsdóttir et al. (2024)	PFOA	HSE	N/A	30	Methanol	NR	0.5	1.07	36	13.5	0.002	3.8E-3

Notes.
APFO = Ammonium Salt of Perfluorooctanoic Acid; ED = Experimental Duration; EL = Experimental Load; FA = Fractional Absorption; HEM = Human Epidermal Membrane; HFT = Full-Thickness Human Skin; HSE = Human Skin Equivalent; J_{ss} = Steady-State Flux; K_p = Dermal Permeability Coefficient; N/A = Not Applicable; NR = Not Reported; PFOA = Perfluorooctanoic Acid; SA = Skin Surface Area.
– = Not Measured.

3.5. Scenario 1: dermally absorbed dose of PFOA from soil

A value of 0.016 (*i.e.*, 1.6% fractional absorption) was used as the ABS_d in Equation (3) to estimate a DA_{Soil} for children. We estimated the DA_{Soil} to be 4.0×10^{-13} mg/cm^2-event. Accordingly, the cumulative DAD_{Soil} for a child was calculated in Equation (4) as 6.0×10^{-11} mg/kg-day. Regarding ingestion, the ADD_{Soil} for a child was calculated based on Equation (7) as 1.6×10^{-9} mg/kg-day. Thus, for children, the dermally absorbed dose of PFOA from soil and dust is approximately 26 times lower than a dose associated with the ingestion of soil and dust. Note that selecting other values for C_{Soil} in Equations 3 and 7 would result in the same estimate of 26-fold lower dermal absorption of PFOA relative to ingestion; that is, the relationship is independent of soil concentration.

3.6. Scenario 2: dermally absorbed dose of PFOA from bathing

A value of 0.000044 cm/h was used as the K_p in Equation (5) to estimate a DA_{Water} for children. Based on the UCMR 3 median and 90th percentile concentrations of PFOA in drinking water, the DA_{Water} was 4.4×10^{-13} and 1.0×10^{-12} mg/cm^2-event, respectively. Accordingly, the DAD_{Water} for children was calculated in Equation (6). Based on the UCMR 3 median and 90th percentile concentrations of PFOA in drinking water, the DAD_{Water} was 1.8×10^{-10} and 4.1×10^{-10} mg/kg-day, respectively. Regarding ingestion, based on the UCMR 3 median and 90th percentile concentrations of PFOA in drinking water, ADD_{Water} values for children were calculated based on Equation (8) as 1.7×10^{-6} and 3.9×10^{-6} mg/kg-day, respectively. Thus, for children, the dermally absorbed dose of PFOA from water *via* bathing is over 9,000 times lower than a dose associated with consuming drinking water. Note that selecting other values for C_{Water} in Equations 5 and 8 would result in the

Table 3
Parameters to calculate N_{derm} for PFOA dermal absorption studies in humans.

Reference	Model	Solvent	pH	EL (μg/cm^2)	K_p (cm/hr)	C_v (μg/cm^3)	J_{ss} ($K_p \times C_v$) (μg/cm^2-hr)	ED (hrs)	Experimental Conditions According to Authors	N_{derm} [EL/(J_{ss} × ED)] (Unitless)	Experimental Conditions Based on N_{derm}
Fasano et al. (2005)	HEM	Water	5.5	46,900	9.5E-7	30	0.190	48	Infinite	5,140	Infinite
Franko et al. (2012)	Study 1: HFT	Acetone	NR	781	–	–	–	24	Finite	–	–
	Study 2: HEM				–	–	–		Finite	–	–
	Study 3: HEM	Water	2.25	2,340	5.5E-2	3,000	165		Infinite	0.592	Finite
	Study 4: HEM		5.01	11	5.8E-5	14	0.0008		Infinite	561	Infinite
	Study 5: HEM	Buffer	5.5	2,340	4.4E-5	3,000	0.132		Infinite	740	Infinite
Abraham and Monien (2022)	Human subject	Sunscreen	NR	0.006	–	3.7	–	48	Not specified	–	–
Ragnarsdóttir et al. (2024)	HSE	Methanol	NR	0.5	3.8E-3	–	0.002	36	Infinite	7.27	Infinite

Notes.
C_v = Concentration in Vehicle; ED = Experimental Duration; EL = Experimental Load; HEM = Human Epidermal Membrane; HFT = Full-Thickness Human Skin; HSE = Human Skin Equivalent; K_p = Dermal Permeability Coefficient; J_{ss} = Steady-State Flux; N_{derm} = Dermal Number; NR = Not Reported; PFOA = Perfluorooctanoic Acid.
– = Not Measured.

Table 4
Strengths and limitations of PFOA dermal absorption kinetic studies for human health risk assessment.

Reference	Fractional Absorption Experimental Measurements		J_{ss} and/or K_p Experimental Measurements	
	Strengths	Limitations	Strengths	Limitations
Fasano et al. (2005)	- Number of subjects - Sample size - Solvent appropriate - pH appropriate	- Infinite dose conditions confirmed based on N_{derm}	- Number of subjects - Sample size - Solvent appropriate - pH appropriate - Infinite dose conditions confirmed based on N_{derm}	- Steady-state conditions not confirmed
Franko et al. (2012)	- None noted (experimental conditions 1 and 2)	- Number of subjects - Sample size - Organic solvent (acetone) - pH not reported - Infinite dose conditions likely based on EL	- Sample size - Solvent appropriate - pH appropriate (experimental conditions 4 and 5) - Infinite dose conditions based on N_{derm} (experimental conditions 4 and 5)	- Number of subjects - pH 2.25 not relevant (experimental condition 3) - Finite dose conditions based on N_{derm} (experimental condition 3) - Steady-state conditions not confirmed
Abraham and Monien (2022)	- Human subject - Relevant exposure conditions - Matrix appropriate - Finite dose conditions likely based on EL	- Single human subject - Sample size - pH of matrix uncertain	- N/A	- N/A
Ragnarsdóttir et al. (2024)	- Sample size	- Organic solvent (methanol) - pH not reported - Infinite dose conditions confirmed based on N_{derm}	- Sample size - Infinite dose conditions confirmed based on N_{derm} - Steady-state conditions confirmed	- Organic solvent (methanol) - pH not reported

Notes.
EL = Experimental Load; J_{ss} = Steady-State Flux; K_p = Dermal Permeability Coefficient; N/A = Not Applicable; N_{derm} = Dermal Number; PFOA = Perfluorooctanoic Acid.

same estimate of 9,000-fold lower dermal absorption of PFOA relative to ingestion; that is, the relationship is independent of water concentration.

4. Discussion

Our evaluation of the four available PFOA dermal absorption kinetic studies identified the most appropriate dermal absorption factors for use in exposure assessments. The reliability and physiological relevance of the reported kinetic measurements were related to the experimental conditions. Below, we provide rationale for the identification of the recommended dermal absorption factors.

4.1. Selection of PFOA fractional absorption value

For fractional absorption experiments, the reliability and physiological relevance of the measurement was most impacted by: (1) EL (*i.e.*, finite dose conditions), and (2) vehicle solvent. We calculated N_{derm} values of 7.27 and 5,140 from the studies by Ragnarsdóttir et al. (2024) and Fasano et al. (2005), respectively, confirming that both experiments were conducted under infinite dose conditions, consistent with the authors' stated intentions. Accordingly, fractional absorption values in these studies were not measured under the appropriate conditions.

Ragnarsdóttir et al. (2024) stated that their infinite dose study evaluated an EL of 0.5 μg/cm^2. We calculated an EL of 46,875 μg/cm^2 for the infinite dose study by Fasano et al. (2005), which likely drove the high calculated value of N_{derm}. Notably, Franko et al. (2012) stated that their fractional absorption experiments were conducted under finite dose conditions. However, we calculated an EL of 781 μg/cm^2, which is three orders of magnitude greater than that of Ragnarsdóttir et al. (2024), indicating that the EL of Franko et al. (2012) should also be considered an infinite dose.

By contrast, we calculated an EL of 0.006 μg/cm^2 for the study by Abraham and Monien (2022), which is three, five, and seven orders of magnitude lower relative to those calculated for Ragnarsdóttir et al. (2024), Fasano et al. (2005), and Franko et al. (2012), respectively. The applied load in the sunscreen was stated to have been "applied on the whole body surface as usually done before a sunbath," (assuming an exposed skin surface area of 19,652 cm^2 for an adult, as recommended by US EPA) (Abraham and Monien, 2022). Ragnarsdóttir et al. (2024) discussed the results of Abraham and Monien (2022) and similarly calculated an EL of 0.0055 μg/cm^2 for that study (assuming an exposed skin surface area of 20,000 cm^2). In contrast, the three *in vitro* studies were conducted in diffusion cell systems with exposed skin surface areas of 0.64 cm^2 (Fasano et al., 2005; Franko et al., 2012) or 1.07 cm^2 (Ragnarsdóttir et al., 2024). Thus, based on the substantially lower EL, it is plausible that Abraham and Monien (2022) measured fractional absorption under finite dose conditions.

Regarding Franko et al. (2012), a limitation was the use of acetone as the solvent. Acetone enhances dermal penetration of amphipathic compounds (Tsai et al., 2001; Chiang et al., 2012), such as PFAS. In addition, in 2021, the Agency for Toxic Substances and Disease Registry (ATSDR) reviewed the results of Franko et al. (2012) and concluded that acetone "may have enhanced PFOA absorption" (ATSDR, 2021). Above, we noted that the ELs applied by Franko et al. (2012) and Fasano et al. (2005) were three and five orders of magnitude greater than that applied by Ragnarsdóttir et al. (2024), respectively. Fractional absorption decreases with increasing applied dose, and thus, one would expect substantially less fractional absorption of PFOA under the conditions of these studies relative to Ragnarsdóttir et al. (2024). Indeed, Fasano et al. (2005) reported fractional absorption of PFOA that was two orders of magnitude lower relative to that reported by Ragnarsdóttir et al. (2024). However, the opposite effect was observed by Franko et al. (2012), that is, fractional absorption was three to five times greater than in the study by Ragnarsdóttir et al. (2024). These results strongly suggest that acetone increased PFOA dermal absorption, and thus, PFOA fractional absorption was overestimated by Franko et al. (2012). We conclude that the fractional absorption measurements by Franko et al. (2012) are less reliable because: (1) based on the EL, the experiments were likely conducted under infinite dose conditions, and (2) the use of acetone solvent greatly increased dermal absorption. We also note that the skin samples were derived from one human subject, with sample sizes of six or seven, and therefore, the extent that the results can be generalized across individuals is uncertain (Law et al., 2020; EC SCCS, 2021).

Regarding Abraham and Monien (2022), the experiment was conducted in a single human subject, and thus, similar to Franko et al. (2012), there is uncertainty whether the results are generalizable across individuals (Law et al., 2020; EC SCCS, 2021). Nevertheless, the experiment has substantial strengths. First, while *in vitro* experiments in

HEM and HSE are informative for estimating dermal absorption potential in humans, Abraham and Monien (2022) conducted a quantitative study in intact skin of a human subject. Second, the authors stated that the experimental concentration of 3.7 μg/g PFOA in the sunscreen was comparable to the results of Fujii et al. (2013), who quantified PFAS in personal care products and reported a maximum of 5.7 μg/g PFOA in a sunscreen product. Thus, the study was representative of a real-world exposure concentration, albeit for a maximum, worst-case scenario, under relevant exposure conditions (*i.e.*, in a sunscreen matrix). Third, there were no indications that the sunscreen matrix, including vehicle solvent and pH, may have altered PFOA dermal bioaccessibility (*i.e.*, the amount of PFOA in a matrix that is soluble and available for absorption). For example, neither acetone nor methanol was present in the sunscreen. The pH of the applied sunscreen was not stated by the authors. However, the pH of sunscreens typically ranges from 4.5 to 6.0 (Matousek et al., 2003; Smaoui et al., 2017; Health Canada, 2020), close to the normal physiological pH of the stratum corneum of 5.5, at which PFOA is almost entirely ionized. It is noteworthy that a recent *in vitro* study concluded that sunscreen had no effect on the dermal bioaccessibility of PFOA (Ragnarsdóttir et al., 2023).

Overall, we conclude that the 1.6% fractional absorption of PFOA reported by Abraham and Monien (2022) is the most appropriate experimental measurement for use as a dermal absorption factor but should be considered provisional due to uncertainty regarding the extent of generalizability across individuals. We note that this provisional value is one order of magnitude below the high-end measurement of 48% reported by Franko et al. (2012), which has been used in some dermal risk assessments. For example, as discussed, Poothong et al. (2020) applied the 48% fractional absorption of PFOA in a dermal risk assessment but acknowledged that this was likely an overestimate. Similarly, the European Union's SCCS recommends a default value of 50% dermal absorption for a chemical in the absence of experimentally determined values (EC SCCS, 2021). The provisional value of 1.6% fractional absorption would be suitable for use (rather than a default assumption of 50%) because it is an experimentally determined value in a human subject that is a conservative representation of real-world exposure conditions (*i.e.*, a finite dose, albeit on the high end of relevance, in a sunscreen matrix).

4.2. Selection of PFOA J_{ss} and K_p values

For J_{ss} and K_p experiments, the reliability and physiological relevance of the measurements were most impacted by (1) EL (*i.e.*, infinite dose conditions), (2) pH, and (3) vehicle solvent. For experimental condition 3 of Franko et al. (2012), we calculated an N_{derm} value of 0.592, suggesting that the experiment was conducted under finite dose conditions. However, this finding is likely an artifact of the experimental pH of 2.25, at which PFOA is expected to exist primarily in a unionized form (the acid dissociation constant [pKa] of PFOA is 3.8). Franko et al. (2012) acknowledged that dermal uptake of PFOA is low at the normal physiological pH of 5.5 of the stratum corneum because PFOA is almost entirely ionized, and therefore has limited capacity to passively cross biological membranes. Franko et al. (2012) also acknowledged that the experimental pH of 2.25 increased the dermal permeation of PFOA and concluded that ionization state is a major consideration for PFOA dermal absorption.

Franko et al. (2012) (with the exception of experimental condition 3) and Fasano et al. (2005) were comparable regarding the number of subjects ($n \geq 3$) and sample size ($n = 8$–10), and both utilized appropriate vehicle solvents (water or citrate buffer) at physiologically relevant pH (5.01–5.5). Although the EL varied by three orders of magnitude in these two experiments, the calculation of the N_{derm} values confirmed that the experiments were conducted appropriately under infinite dose conditions, and therefore likely reliably measured J_{ss} and K_p.

For the study conducted by Ragnarsdóttir et al. (2024), we calculated an N_{derm} value of 7.27, suggesting that J_{ss} and K_p were also measured appropriately under infinite dose conditions. A substantial strength of this study relative to Fasano et al. (2005) and Franko et al. (2012) is that Ragnarsdóttir et al. (2024) provided data confirming that steady-state conditions were achieved during experimental measurements of J_{ss} and K_p. Ragnarsdóttir et al. (2024) noted that their measured value of K_p was 10-fold lower than that measured for unionized PFOA in experimental condition 3 of Franko et al. (2012) and stated that the difference could be attributed to differences in vehicle solvent, skin barrier, exposure time, and dose. Nevertheless, the authors did not report the pH of the PFOA formulation, so the ionization state of PFOA in their experiment was unclear. In addition, the authors did not evaluate the effect of the methanol solvent on skin surface pH or the magnitude of its effect on the dermal permeability of the 3D-HSE. Accordingly, Ragnarsdóttir et al. (2024) concluded that a limitation of their study was that the effect of the methanol solvent could not be completely excluded as a factor impacting the dermal uptake of PFAS. Thus, we conclude that the J_{ss} and K_p measurements by Ragnarsdóttir et al. (2024) were appropriately measured under infinite dose conditions but have uncertain physiological relevance because: (1) the ionization state of PFOA in this experiment was unknown (*i.e.*, pH was not reported), and (2) as stated by the authors, the impact of the methanol solvent on dermal uptake was unknown.

Overall, we conclude that 0.132 μg/cm^2-hr and 0.000044 cm/h for J_{ss} and K_p, respectively, as measured by Franko et al. (2012), are the most physiologically relevant experimental measurements for use in human exposure assessments but should be considered provisional until more data are available.

4.3. Dermally absorbed dose of PFOA from soil and water

Our provisional recommendations of 1.6% fractional absorption and 0.000044 cm/h for the K_p for PFOA are similar to, or lower than, the lowest respective values identified for other contaminants, indicating that PFOA has low dermal absorption potential relative to other environmental contaminants in soil and water (US EPA, 2004). Regarding soil, US EPA (2004) has recommended dermal fractional absorption values for a variety of compounds (based on experimental means). Our provisional recommendation of 1.6% fractional absorption of PFOA as reported by Abraham and Monien (2022) is similar to the lowest soil values identified by US EPA (2004), specifically 0.1% for cadmium and 3% for dichloro-diphenyl-trichloroethane (DDT), 2,3,7,8-tetrachlorodibenzo-p-dioxin (TCDD) and other dioxins, and arsenic. Regarding dermal exposures from water, US EPA (2004) has recommended K_p values for inorganic and organic compounds (based on modeled predictions or experimental values). Our provisional recommendation of 0.000044 cm/h for the K_p for PFOA as measured by Franko et al. (2012) is one to four orders of magnitude lower than experimental values reported for inorganic compounds (ranging from 0.0001 cm/h for lead and mercury [+2] to 0.24 cm/h for mercury vapor [US EPA, 2004]). For organic compounds, US EPA has recommended using its predicted K_p values to ensure consistency between agency risk assessments (US EPA, 2004). The predicted K_p values ranged from 0.00001 cm/h for tris (aziridinyl)-para-benzo quinone to 1.5 cm/h for dibenzo (a,h)anthracene (US EPA, 2004). Thus, our provisional recommendation of 0.000044 cm/h for the K_p for PFOA is on the same order of magnitude as the lowest values of K_p for organic compounds recommended by US EPA (2004). These findings are consistent with the recommendations of scientific and regulatory agencies that, for most PFAS, showering or bathing in water containing PFAS should not be an important contributor to increased exposures, particularly relative to ingestion of drinking water (ATSDR, 2024; US EPA, 2024b).

Abraham and Monien (2022) reported a body burden of 132 ng/L PFOA 22 days after a single application of PFOA in sunscreen, representing 9.4% of the human volunteer's background exposure level (1400 ng/L). Hence, the authors suggested that, given repeated exposures at sufficiently high concentrations, dermal exposures to PFAS may

contribute to PFAS body burdens. Overall, however, our analyses determined that absorbed doses of PFOA from soil and water for children were over 20–9,000 times lower *via* dermal exposure relative to ingestion, respectively. The substantially lower doses estimated for the dermal route relative to ingestion are consistent with the general understanding that for PFAS with physicochemical characteristics similar to PFOA (discussed further in the following section), dermal absorption is a potential, but minor, exposure pathway for humans relative to ingestion (Haug et al., 2011; Sunderland et al., 2019; Poothong et al., 2020). Finally, though outside the scope of our evaluation, we note that showering and bathing may also involve inhalation exposure, and recent studies suggest the potential for inhalation exposure to neutral and volatile PFAS (Morales-McDevitt et al., 2021; Kissel et al., 2023). We did not identify any information regarding potential inhalation exposure to PFOA (an ionic PFAS) from showering or bathing, precluding a comparison to estimated dermal and ingestion exposures. However, we note that PFOA is ionized in water, so is essentially nonvolatile from water (Barton et al., 2007; Health Canada, 2018).

4.4. Determinants of dermal absorption for PFOA and other PFAS

The physicochemical properties of PFAS determine the extent of their dermal absorption. As discussed, Franko et al. (2012) concluded that the low dermal absorption of PFOA under environmentally relevant exposure conditions is due, in part, to its ionized status (*i.e.*, pKa = 3.8). Similarly, Ragnarsdóttir et al. (2024) evaluated 17 PFAS (10 PFCAs, including PFOA, and 7 PFSAs) and observed decreasing fractional absorption with increasing carbon chain length and log octanol-water partition coefficient (K_{ow}). The authors also observed greater fractional absorption for the 10 PFCAs relative to the 7 PFSAs, and for PFCAs relative to PFSAs with the same carbon chain length, suggesting that functional group composition also plays a role in dermal absorption. Hence, although almost all experimental measurements of the dermal absorption kinetics of PFAS in human skin to date are for PFOA, these data can provide information about the dermal absorption potential of other PFAS that share similar physicochemical properties (*e.g.*, pKa, carbon chain length, functional group composition, log K_{ow}). That is, PFAS that have physicochemical properties similar to those of PFOA would be expected to exhibit similarly low dermal absorption in human skin under environmentally relevant conditions.

Finally, based on the effect of vehicle solvents on PFOA dermal absorption, further discussion is warranted regarding the sunscreen matrix evaluated by Abraham and Monien (2022). To increase the dermal absorption of lipophilic or hydrophilic active compounds, cosmetics may contain ingredients that serve as dermal penetration enhancers, including water, alcohols, surfactants, fatty acids, or terpenes and terpenoids (Kim et al., 2020), as well as PFAS themselves (Fujii et al., 2013). Abraham and Monien (2022) provided the full list of ingredients of the evaluated commercial sunscreen, but it is unclear if any of the ingredients served as dermal penetration enhancers. As mentioned, however, the available studies provide no evidence that sunscreen matrices alter the dermal bioaccessibility of PFOA (Abraham and Monien, 2022; Ragnarsdóttir et al., 2023).

5. Conclusions

Our evaluation of experimental measurements of PFOA dermal absorption kinetics determined that the criteria that had the greatest impacts on measurement reliability and physiological relevance were EL, solvent, and pH. Due to the limited number of studies of PFOA dermal absorption kinetics, the recommended dermal absorption factors should be considered provisional for application in exposure assessments until more data are available. Nevertheless, based on the recommended dermal absorption factors we demonstrated that dermal absorption of PFOA is several orders of magnitude lower than levels *via* ingestion. Finally, low dermal absorption would be expected for PFAS with similar chemistry and structure to PFOA (*e.g.*, pKa, carbon chain length, functional group composition, log K_{ow}). Overall, we conclude that dermal absorption of PFOA is low under physiologically and environmentally relevant conditions (*e.g.*, with respect to matrix, pH, and finite dose conditions). Our results are consistent with the general understanding that the dermal exposure pathway is minor relative to the ingestion pathway for PFOA.

Funding

The preparation of this manuscript was supported by the authors' employer (Gradient). Some of the underlying research was funded by a defendant involved in litigation regarding PFAS.

CRediT authorship contribution statement

Andrew Yeh: Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Robyn L. Prueitt:** Writing – review & editing, Validation, Formal analysis, Conceptualization. **Laura E. Kerper:** Writing – review & editing, Validation, Investigation, Formal analysis. **Barbara D. Beck:** Writing – review & editing, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

Gradient has been involved in several litigation and non-litigation matters involving PFAS. Drs. Barbara Beck and Robyn L. Prueitt have served as experts in litigation matters involving PFAS. Some of the underlying research and analysis for this paper was performed during the context of those engagements and was sponsored by the defendant. Neither the law firms nor the defendant in those matters, however, asked that this paper be written or published. The preparation of this manuscript was supported only by the authors' employer (Gradient), and its conclusions are exclusively those of the authors. Aside from the authors and internal Gradient reviewers, no one has commented on or revised this manuscript prior to its submission.

Acknowledgements

We thank Jiayang Chien, Rosemary Mattuck, Hilary J. Robinson, and William Dowd for technical support, and Laura E. Smith for editorial support.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.yrtph.2024.105766.

Data availability

No data was used for the research described in the article.

References

Abraham, K., Monien, B.H., 2022. Transdermal absorption of 13C4-perfluorooctanoic acid (13C4-PFOA) from a sunscreen in a male volunteer - what could be the contribution of cosmetics to the internal exposure of perfluoroalkyl substances (PFAS)? Environ. Int. 169, 107549. https://doi.org/10.1016/j.envint.2022.107549.

Agency for Toxic Substances and Disease Registry (ATSDR), 2021. "Toxicological Profile for Perfluoroalkyls.", p. 993. May.

Agency for Toxic Substances and Disease Registry (ATSDR), 2024. "Preventing PFAS exposure.". November 12. https://www.atsdr.cdc.gov/pfas/prevent-exposure/your-body.html#:~:text=Research/20has/20suggested/20that/20exposure,exposure/20from/20contaminated/20drinking/20water.

Barton, C.A., Kaiser, M.A., Russell, M.H., 2007. "Partitioning and removal of perfluorooctanoate during rain events: the importance of physical-chemical properties.". J. Environ. Monit. 9 (8), 839–846. https://doi.org/10.1039/b703510a.

Chen, Q., Yi, S., Ye, Q., Zhu, Y., Zhong, W., Zhu, L., 2022. "Insights into the dermal absorption, deposition, and elimination of poly- and perfluoroalkyl substances in rats: the importance of skin exposure.". Environ. Sci. Technol. 56 (23), 16975–16984. https://doi.org/10.1021/acs.est.2c03181.

Chiang, A., Tudela, E., Maibach, H.I., 2012. "Percutaneous absorption in diseased skin: an overview.". J. Appl. Toxicol. 32 (8), 537–563. https://doi.org/10.1002/jat.1773.

European Commission, Health & Consumer Protection Directorate-General, Scientific Committee on Consumer Safety (EC SCCS), 2021. "The SCCS Notes of Guidance for the Testing of Cosmetic Ingredients and Their Safety Evaluation (11th Revision).". SCCS/1628/21. 194pp.

European Food Safety Authority (EFSA), 2017. "Guidance on dermal absorption.". EFSA J. 15 (6), e04873. https://doi.org/10.2903/j.efsa.2017.4873.

Fasano, W.J., Kennedy, G.L., Szostek, B., Farrar, D.G., Ward, R.J., Haroun, L., Hinderliter, P.M., 2005. Penetration of ammonium perfluorooctanoate through rat and human skin in vitro. Drug Chem. Toxicol. 28 (1), 79–90. https://doi.org/10.1081/DCT-39707.

Franko, J., Meade, B.J., Frasch, H.F., Barbero, A.M., Anderson, S.E., 2012. Dermal penetration potential of perfluorooctanoic acid (PFOA) in human and mouse skin. J. Toxicol. Environ. Health 75 (1), 50–62. https://doi.org/10.1080/15287394.2011.615108.

Frasch, H.F., Dotson, G.S., Bunge, A.L., Chen, C.P., Cherrie, J.W., Kasting, G.B., Kissel, J. C., Sahmel, J., Semple, S., Wilkinson, S., 2014. Analysis of finite dose dermal absorption data: implications for dermal exposure assessment. J. Expo. Sci. Environ. Epidemiol. 24 (1), 65–73. https://doi.org/10.1038/jes.2013.23.

Fujii, Y., Harada, K.H., Koizumi, A., 2013. "Occurrence of perfluorinated carboxylic acids (PFCAs) in personal care products and compounding agents.". Chemosphere 93 (3), 538–544. https://doi.org/10.1016/j.chemosphere.2013.06.049.

Haug, L.S., Huber, S., Becher, G., Thomsen, C., 2011. Characterisation of human exposure pathways to perfluorinated compounds – comparing exposure estimates with biomarkers of exposure. Environ. Int. 37 (4), 687–693. https://doi.org/10.1016/j.envint.2011.01.011.

Health Canada, 2018. "Guidelines for Canadian Drinking Water Quality: Perfluorooctanoic Acid (PFOA).", p. 120p. December.

Health Canada, 2020. 2018 report on compliance monitoring: sunscreen testing. September 10. https://www.canada.ca/en/health-canada/services/publications/drugs-health-products/. (Accessed 31 May 2024).

Kim, B., Cho, H., Moon, S., Ahn, H., Bae, S., Cho, H., An, S., 2020. "Transdermal delivery systems in cosmetics. " Biomed. Dermatol 4, 10. https://doi.org/10.1186/s41702-020-0058-7.

Kissel, J.C., 2011. The mismeasure of dermal absorption. J. Expo. Sci. Environ. Epidemiol. 21 (3), 302–309. https://doi.org/10.1038/jes.2010.22.

Kissel, J.C., Titaley, I.A., Muensterman, D.J., Field, J.A., 2023. Evaluating neutral PFAS for potential dermal absorption from the gas phase. Environ. Sci. Technol. 57 (12), 4951–4958. https://doi.org/10.1021/acs.est.2c08835.

Law, R.M., Ngo, M.A., Maibach, H.I., 2020. "Twenty clinically pertinent factors/observations for percutaneous absorption in humans.". Am. J. Clin. Dermatol. 21 (1), 85–95. https://doi.org/10.1007/s40257-019-00480-4.

Matousek, J.L., Campbell, K.L., Kakoma, I., Solter, P.F., Schaeffer, D.J., 2003. "Evaluation of the effect of pH on *in vitro* growth of *Malassezia pachyd*ermatis.". Can. J. Vet. Res. 67 (1), 56–59.

Morales-McDevitt, M.E., Becanova, J., Blum, A., Bruton, T.A., Vojta, S., Woodward, M., Lohmann, R., 2021. "The air that we breathe: neutral and volatile PFAS in indoor air.". Environ. Sci. Technol. Lett. 8 (10), 897–902. https://doi.org/10.1021/acs.estlett.1c00481.

Organisation for Economic Co-operation and Development (OECD), 2022. Guidance Notes on Dermal Absorption Studies, second ed. ENV/JM/MONO(2011)36/REV1, OECD Series on Testing and Assessment No. 156. 61pp.

Poothong, S., Papadopoulou, E., Padilla-Sanchez, J.A., Thomsen, C., Haug, L.S., 2020. Multiple pathways of human exposure to poly- and perfluoroalkyl substances (PFASs): from external exposure to human blood. Environ. Int. 134, 105244. https://doi.org/10.1016/j.envint.2019.105244.

Ragnarsdóttir, O., Abdallah, M.A., Harrad, S., 2022. Dermal uptake: an important pathway of human exposure to perfluoroalkyl substances? Environ. Pollut. p119478. https://doi.org/10.1016/j.envpol.2022.119478.

Ragnarsdóttir, O., Abdallah, M.A., Harrad, S., 2023. Dermal bioaccessibility of perfluoroalkyl substances from household dust; Influence of topically applied cosmetics. Environ. Res. 238 (Pt. 1), 117093. https://doi.org/10.1016/j.envres.2023.117093.

Ragnarsdóttir, O., Abou-Elwafa Abdallah, M., Harrad, S., 2024. "Dermal bioavailability of perfluoroalkyl substances using in vitro 3D human skin equivalent models.". Environ. Int. 188, 108772. https://doi.org/10.1016/j.envint.2024.108772.

Smaoui, S., Ben Hlima, H., Ben Chobba, I., Kadri, A., 2017. "Development and stability studies of sunscreen cream formulations containing three photo-protective filters.". Arab. J. Chem. 10 (Suppl. 1), S1216–S1222. https://doi.org/10.1016/j.arabjc.2013.02.020.

Stalcup, D., 2014. US EPA, Office of Solid Waste and Emergency Response (OSWER). Memo. Superfund Nat. Pol. Managers, Regions 1-10 re: Hum. Health Eval. Manual, Suppl. Guid.: Update Stand. Default Expo Factor. OSWER Dir. 9200, 1 120. 7pp., February 6. https://www.epa.gov/sites/default/files/2015-11/documents/oswer_directive_9200.1-120_exposurefactors_corrected2.pdf.

Strynar, M.J., Lindstrom, A.B., Nakayama, S.F., Egeghy, P.P., Helfant, L.J., 2012. Pilot scale application of a method for the analysis of perfluorinated compounds in surface soils. Chemosphere 86 (3), 252–257. https://doi.org/10.1016/j.chemosphere.2011.09.036.

Sunderland, E.M., Hu, X.C., Dassuncao, C., Tokranov, A.K., Wagner, C.C., Allen, J.G., 2019. "A review of the pathways of human exposure to poly- and perfluoroalkyl substances (PFASs) and present understanding of health effects.". J. Expo. Sci. Environ. Epidemiol. 29 (2), 131–147. https://doi.org/10.1038/s41370-018-0094-1.

Tsai, J.C., Sheu, H.M., Hung, P.L., Cheng, C.L., 2001. "Effect of barrier disruption by acetone treatment on the permeability of compounds with various lipophilicities: implications for the permeability of compromised skin.". J. Pharmaceut. Sci. 90 (9), 1242–1254. https://doi.org/10.1002/jps.1077.

US EPA, 2004. Risk Assessment guidance for superfund (RAGS). Volume I: Human Health Evaluation Manual (Part E, Supplemental Guidance for Dermal Risk Assessment) (Final). Off. Superfund Remediat. Technol. Innovat. EPA/540/R/99/005, OSWER 9285.7-02EP, PB99-963312. 156pp., July.

US EPA, 2016. Health Effects Support Document for Perfluorooctanoic Acid (PFOA). Office of Water. EPA 822-R-16-003. 322pp., May.

US EPA, 2017a. "Update for Chapter 5 of the Exposure Factors Handbook: Soil and Dust Ingestion.". Office of Research and Development, National Center for Environmental Assessment (NCEA). EPA/600/R-17/384F. 100pp., September.

US EPA, 2017b. Third unregulated contaminant monitoring Rule (UCMR 3) occurrence data. Off. Water, January. Accessed at. https://www.epa.gov/dwucmr/occurrence-data-unregulated-contaminant-monitoring-rule.

US EPA, 2019. Update for Chapter 3 of the Exposure Factors Handbook: Ingestion of Water and Other Select Liquids." Office of Research and Development. National Center for Environmental Assessment (NCEA). EPA/600/R-18/259F. 157pp., February.

US EPA, 2022. Interim Drinking Water Health Advisory: Perfluorooctanoic Acid (PFOA) (CASRN 335-67-1). Office of Water, Office of Science and Technology, Health and Ecological Criteria Division. EPA/822/R-22/003. 33pp., June.

US EPA, 2024a. "Exposure assessment tools by routes - ingestion.". January 2. https://www.epa.gov/expobox/exposure-assessment-tools-routes-ingestion.

US EPA, 2024b. "PFAS explained: meaningful and achievable steps you can take to reduce your risk.". November 21. https://www.epa.gov/pfas/meaningful-and-achievable-steps-you-can-take-reduce-your-risk.